

Hotel Booking Cancellation Analysis

Business Problem

Hotel reservation cancellations have become a significant challenge for both City Hotels and Resort Hotels, affecting revenue generation, occupancy rates, and operational planning. High cancellation volumes can result in unoccupied rooms, inefficient resource utilization, and difficulties in forecasting demand.

The objective of this analysis is to investigate booking cancellation patterns, identify the factors contributing to reservation cancellations, and provide data driven recommendations that can help hotels improve customer retention, optimize pricing strategies, and enhance overall business performance.

Assumptions

1. No extraordinary events between 2015 and 2017 significantly influenced customer booking behavior within the dataset.
2. The booking information available in the dataset accurately represents hotel reservation patterns and can be used for meaningful analysis.
3. The recommended business strategies can be implemented without causing adverse effects on hotel operations.
4. The hotels are not currently applying the recommendations proposed in this report.
5. Reservation cancellations have a measurable impact on hotel revenue and occupancy management.
6. Cancelled reservations contribute to vacant room inventory during the intended booking period.
7. Most customers complete their booking and cancellation activities within the same calendar year.

Research Questions

1. Which factors have the greatest influence on hotel reservation cancellations?
2. What strategies can hotels adopt to reduce cancellation rates and improve booking retention?
3. How can pricing, marketing, and booking channel analysis support better business decision making?

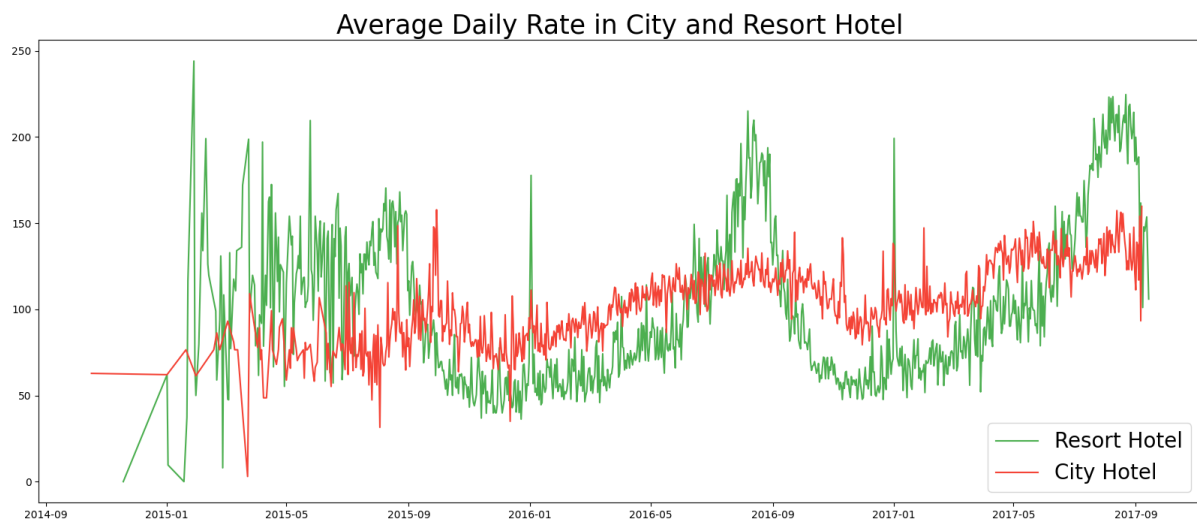
Assumption

1. Reservations with higher room rates are more likely to be cancelled than lower priced bookings.
2. Seasonal trends and booking periods influence the likelihood of reservation cancellations.
3. Online Travel Agencies (OTAs) contribute a larger share of reservations and cancellations compared to other booking channels.
4. Cancellation behavior varies across customer locations and geographic regions.

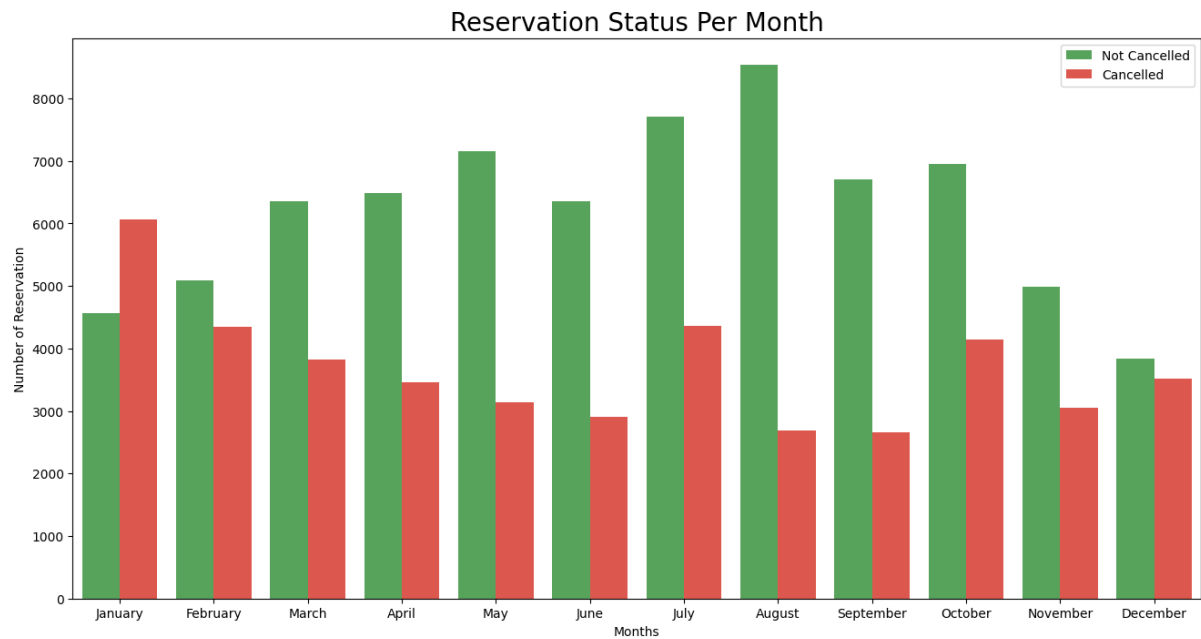
Analysis and Findings

The bar chart illustrates the proportion of cancelled and non cancelled reservations. It can be observed that the majority of bookings were successfully completed. However, approximately 37% of reservations were cancelled, representing a considerable share of total bookings and potentially impacting hotel revenue and occupancy levels.

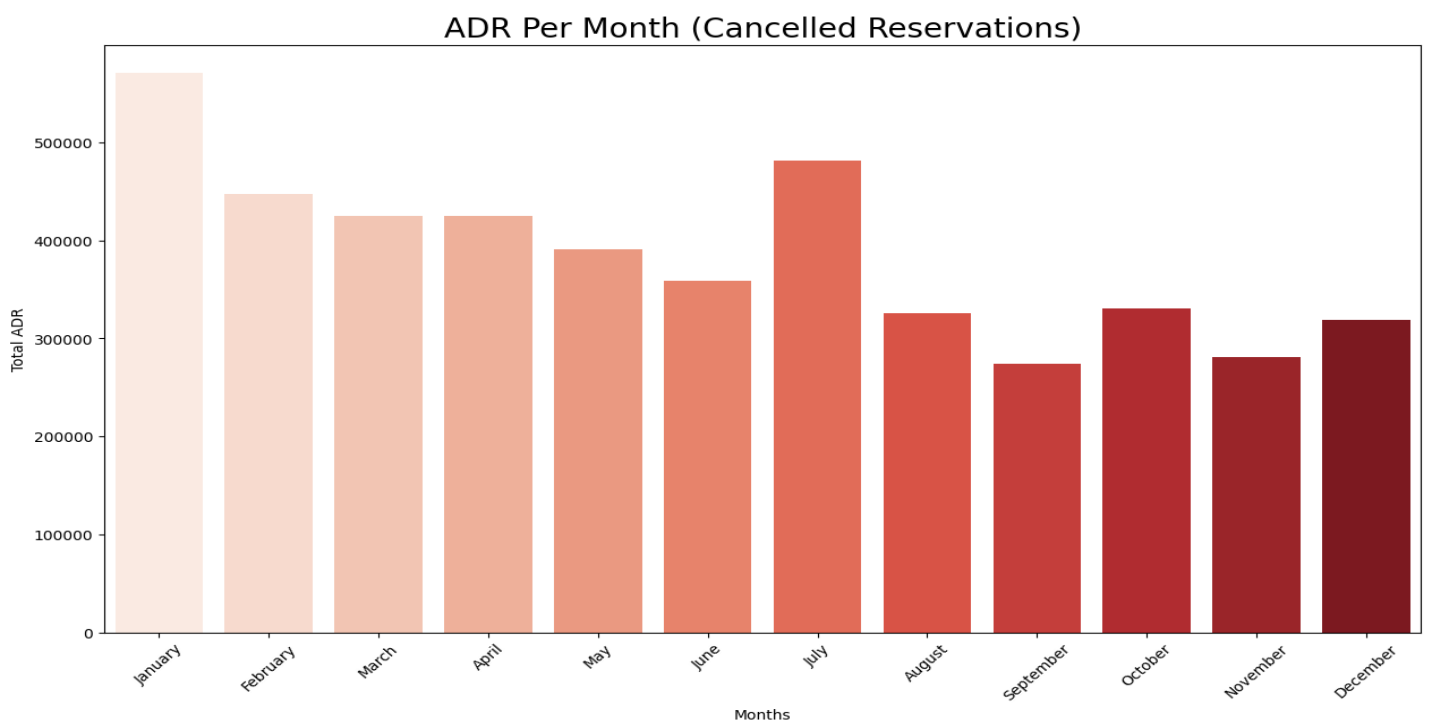
The analysis also shows that City Hotels received a higher number of bookings compared to Resort Hotels. This difference may be due to higher demand for city accommodations from both business and leisure travellers throughout the year.



The line graph comparing Average Daily Rate (ADR) indicates that room prices vary between City Hotels and Resort Hotels over time. On some days, Resort Hotels recorded higher ADR values, while on others, City Hotels had higher average room rates. These fluctuations may be influenced by seasonal demand, weekends, holidays, and other travel related factors.

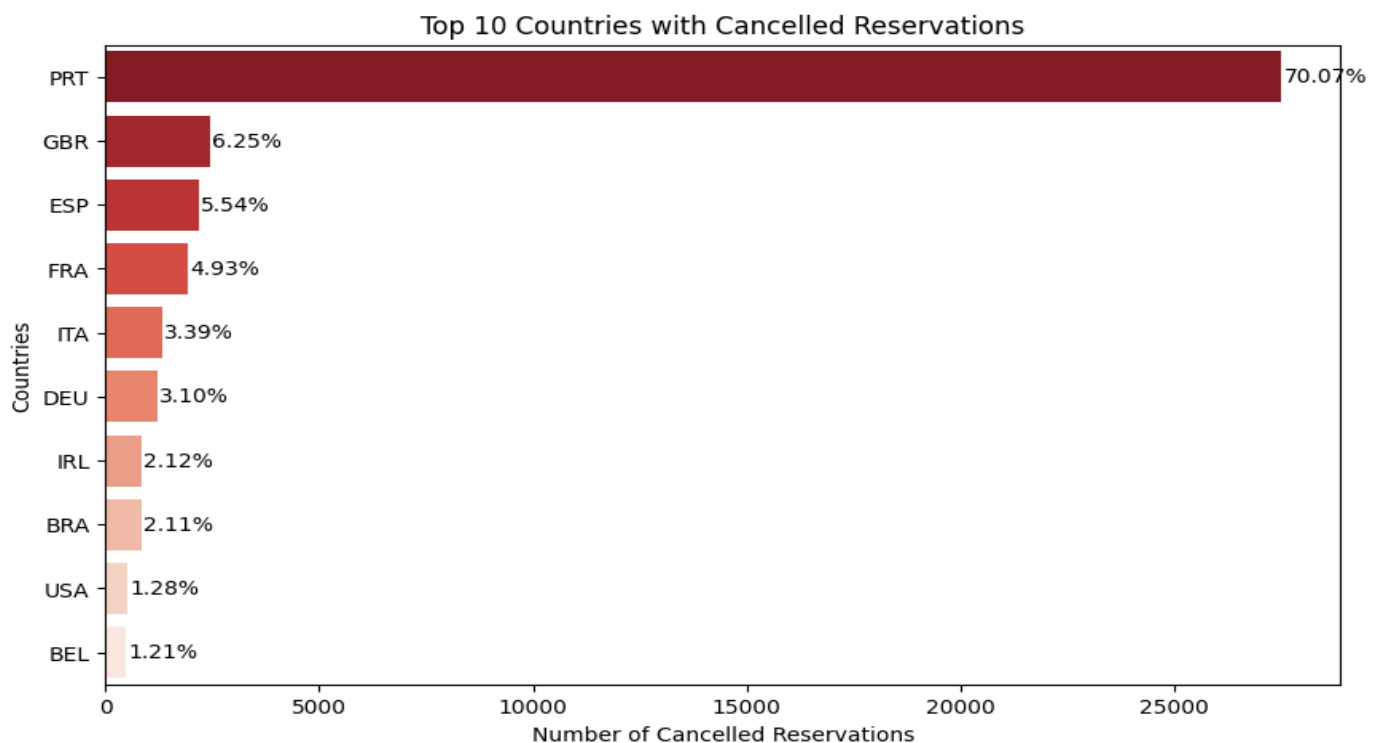


The grouped bar chart was created to examine monthly reservation trends based on booking status. The analysis shows that August recorded the highest number of overall reservations, including both confirmed and cancelled bookings. In contrast, January exhibited the highest cancellation rate relative to the number of reservations made during that month. These findings suggest that reservation activity and cancellation behavior vary significantly across different periods of the year.

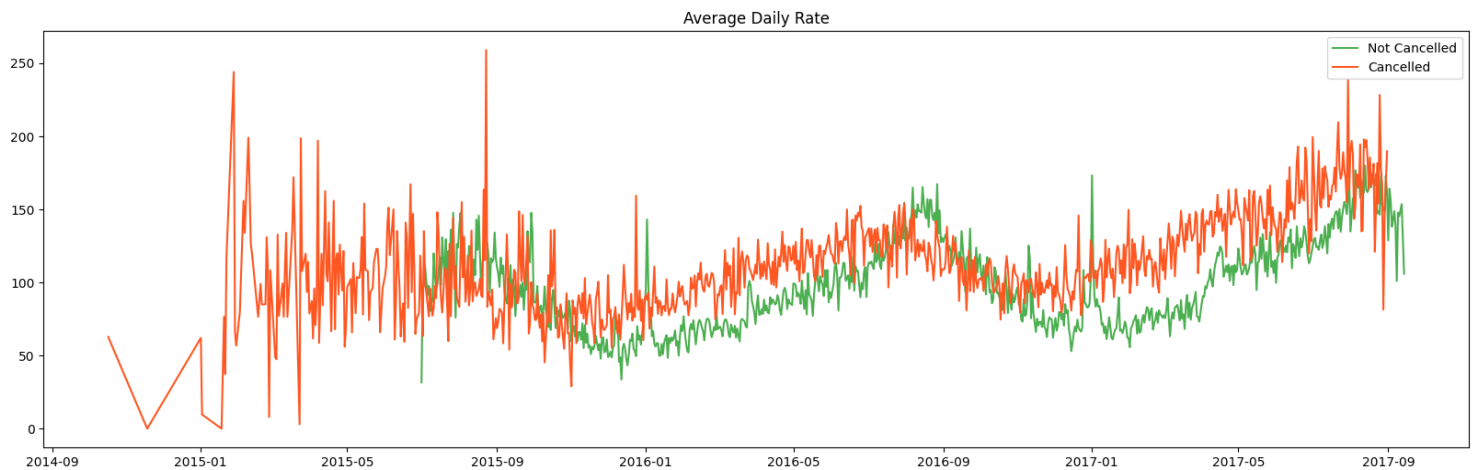


The bar chart indicates a relationship between room prices and reservation cancellations. Higher ADR (Average Daily Rate) values are generally associated with a greater number of booking cancellations, while lower room rates tend to have fewer cancellations. This suggests that pricing is an important factor influencing customer cancellation behavior, although other factors may also contribute to reservation decisions.

To further understand the geographical distribution of booking cancellations, a country wise analysis was conducted. The results indicate that Portugal recorded the highest number of cancelled reservations among all countries in the dataset. This suggests that a significant proportion of cancellation activity originated from Portuguese customers, making it an important market for further investigation and targeted customer retention strategies.



To better understand customer booking behavior, the distribution of reservations across different market segments was analysed. The results show that Online Travel Agencies (OTAs) were the primary source of hotel bookings, accounting for approximately 46% of all reservations. Group bookings represented around 27% of total reservations, making them the second largest segment. In contrast, direct bookings contributed only about 4% of reservations, indicating that most customers preferred using third party booking platforms rather than booking directly with the hotels.



The ADR analysis indicates that cancelled reservations generally have higher average daily rates compared to successfully completed bookings. This observation supports the earlier findings that room pricing plays an important role in customer cancellation behavior. As room rates increase, the likelihood of reservation cancellations also tends to rise.

Recommendations

1. Since higher room prices are associated with increased cancellation rates, hotels should consider adopting flexible pricing strategies. Offering targeted discounts and promotional packages during high price periods may help improve booking retention.
2. Resort Hotels experience a relatively higher proportion of cancellations compared to City Hotels. Introducing special weekend and holiday offers, along with competitive room pricing, could encourage guests to maintain their reservations.
3. As January records the highest cancellation activity, hotels should implement targeted marketing campaigns and customer engagement initiatives during this period to improve occupancy and revenue generation.
4. Given that Portugal contributes the highest number of cancelled reservations, hotel management should focus on enhancing customer experience, service quality, and retention strategies within this market to reduce future cancellation rates.